

# CA5 Question 1 Report

## Legal Retrieval Pipeline

tahamajs

Student ID: 810101504

February 14, 2026

## 1 Project Objective

Question 1 targets a practical Persian legal retrieval pipeline that supports evidence-grounded responses on Iranian legal texts. The final implementation emphasizes:

- deterministic preprocessing and indexing,
- transparent node-level latency tracking,
- retrieval traceability through source snippets,
- reproducible artifact generation for report delivery.

## 2 Implementation Scope

The Q1 implementation is modularized under `answers/Q1/src` and orchestrated by `answers/scripts/run_q1.py`. Core modules are:

- `q1_pipeline.py`: loading, OCR fallback, chunking, TF-IDF indexing, retrieval, reranking, response generation,
- `q1_eval.py`: benchmark execution and RAGAS evaluation export,
- `q1_plots.py`: timing and evaluation plot generation.

## 3 Implementation Outputs

| Artifact                                  | Description                                                                    |
|-------------------------------------------|--------------------------------------------------------------------------------|
| <code>Q1/data/indexed_dataset.pkl</code>  | Serialized TF-IDF index and retrieval metadata                                 |
| <code>Q1/data/pipeline_timings.csv</code> | End-to-end timing, node timings, and retrieval diagnostics per benchmark query |

|                                                                    |                                                                                                                                        |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Q1/data/<br>rag_evaluation_report.xlsx<br>reports/figures/q1/*.png | RAGAS export with <code>faithfulness</code> and<br><code>answer_relevancy</code> columns<br>Plot set used directly in the final report |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|

## 4 Artifact Integrity Snapshot

The artifact snapshot below reflects the current local run state on February 14, 2026.

| Artifact                               | Size (bytes) | Last modified (local time) |
|----------------------------------------|--------------|----------------------------|
| Q1/data/indexed_dataset.pkl            | 9289627      | 2026-02-13 12:25:09        |
| Q1/data/pipeline_timings.csv           | 49673        | 2026-02-13 12:25:23        |
| Q1/data/<br>rag_evaluation_report.xlsx | 16896        | 2026-02-13 10:05:26        |
| reports/figures/q1/*.png               | 6 files      | 2026-02-13 generation set  |

## 5 Data and Indexing Summary

The indexed corpus characteristics from `indexed_dataset.pkl` are:

- source PDF files: 7,
- indexed chunks: 444,
- chunk length mean: 1187.93 characters,
- chunk length range: 165 to 1200 characters,
- chunk length standard deviation: 95.86 characters,
- extracted year values (unique): 51,
- chunks with non-null extracted year: 323 of 444 (72.75%).

### Corpus distribution by source law file

| Source file (English descriptor)              | Chunks | Share  |
|-----------------------------------------------|--------|--------|
| Law on combating goods and currency smuggling | 98     | 22.07% |
| Labor law                                     | 87     | 19.59% |
| Law on registration of deeds and properties   | 86     | 19.37% |
| Value added tax law                           | 77     | 17.34% |
| Social security law                           | 68     | 15.32% |
| Law on issuance of cheques                    | 23     | 5.18%  |
| Landlord and tenant relations law             | 5      | 1.13%  |

## 6 Pipeline Architecture

The execution graph is:

1. dataset loading and OCR-aware text extraction,
2. chunking and metadata assignment,
3. TF-IDF vectorization and index persistence,
4. query rewrite, intent classification, metadata extraction,
5. top- $k$  retrieval with cosine similarity,
6. reranking to top- $n$  evidence blocks,
7. grounded answer synthesis with cited snippets,
8. node-level latency logging and benchmark export.

## 7 Benchmark Configuration

The benchmark currently includes 5 legal questions. All records in the latest run were classified as `law_question`, and all runs returned 3 contexts per query.

### Benchmark case inventory

| ID | Theme                              | Legal area     | Observed latency |
|----|------------------------------------|----------------|------------------|
| 1  | Labor law overtime constraints     | Labor law      | Very high        |
| 2  | Bounced cheque issuance conditions | Commercial law | Low              |
| 3  | Civil law lease rules              | Civil law      | Low              |
| 4  | Tax law offense coverage           | Tax law        | Low              |
| 5  | Social security legal protections  | Social law     | Moderate         |

Ground-truth answer text is not yet populated in the benchmark file (`ground_truth` is null for all 5 rows), so this run focuses on retrieval consistency and runtime behavior rather than reference-based answer scoring.

## 8 Latency Results

### Aggregate latency

| Metric                | Value (seconds) |
|-----------------------|-----------------|
| Total latency mean    | 0.3474          |
| Total latency median  | 0.0809          |
| Total latency minimum | 0.0754          |

|                       |        |
|-----------------------|--------|
| Total latency maximum | 1.3964 |
|-----------------------|--------|

## Node-level mean latency

| Node             | Mean (seconds) |
|------------------|----------------|
| rewrite          | 0.0028         |
| classify_intent  | 0.0033         |
| extract_metadata | 0.0235         |
| context_retrieve | 0.3141         |
| rerank           | 0.0007         |
| generate_answer  | 0.0021         |

## Per-query latency

| Benchmark query theme              | Total latency (seconds) |
|------------------------------------|-------------------------|
| Labor law overtime constraints     | 1.3964                  |
| Bounced cheque issuance conditions | 0.0778                  |
| Civil law lease rules              | 0.0754                  |
| Tax law offense coverage           | 0.0809                  |
| Social security legal protections  | 0.1065                  |

## Stability diagnostics

| Metric                                           | Value          |
|--------------------------------------------------|----------------|
| Standard deviation of total latency              | 0.5866 seconds |
| Coefficient of variation of total latency        | 1.6884         |
| Mean retrieval-stage share of end-to-end latency | 90.40%         |

## Tail-latency diagnostics

| Metric                      | Value          |
|-----------------------------|----------------|
| P90 total latency           | 0.8805 seconds |
| P95 total latency           | 1.1385 seconds |
| Interquartile range (Q3-Q1) | 0.0287 seconds |

## Node contribution to total latency

| Node             | Mean latency (s) | Share of total mean |
|------------------|------------------|---------------------|
| rewrite          | 0.0028           | 0.81%               |
| classify_intent  | 0.0033           | 0.96%               |
| extract_metadata | 0.0235           | 6.76%               |
| context_retrieve | 0.3141           | 90.40%              |
| rerank           | 0.0007           | 0.21%               |
| generate_answer  | 0.0021           | 0.61%               |

## 9 Retrieval Effectiveness Diagnostics

| Diagnostic                             | Value |
|----------------------------------------|-------|
| Hit@1 mean                             | 1.00  |
| Hit@3 mean                             | 1.00  |
| Mean contexts returned per query       | 3.00  |
| Rows with non-empty answer text        | 5 / 5 |
| Rows with non-empty retrieved contexts | 5 / 5 |

### Retrieved-score diagnostics

| Metric                         | Value            |
|--------------------------------|------------------|
| Top-1 similarity score mean    | 0.1770           |
| Top-1 similarity score range   | 0.1123 to 0.2385 |
| Top-3 similarity score mean    | 0.1273           |
| Top-1 minus Top-3 spread mean  | 0.0497           |
| Top-1 minus Top-3 spread range | 0.0035 to 0.0952 |

The retrieval diagnostics indicate complete benchmark recall for this five-query sample. Because the sample size is small, the result should be interpreted as a functional validation signal, not a final quality bound.

## 10 Evaluation Results

RAGAS is configured with `faithfulness` and `answer_relevancy`. The latest live run produced no valid metric values because the OpenAI credential failed authentication during evaluation. In `rag_evaluation_report.xlsx`, both metric columns are null for all 5 rows.

Despite that failure mode, the run confirms the evaluation export path is wired correctly and that question-answer-context tuples are persisted in a format ready for immediate re-evaluation after credential correction. The strict preflight command executed on February 14, 2026 returned: OpenAI API credential check failed with HTTP status 401.

## 11 Operational Quality Notes

- The index and timing artifacts are complete and internally consistent with the benchmark row count.
- The dominant bottleneck is retrieval, not query rewriting or final response assembly.
- High latency variance is concentrated in one broad legal query; four queries are below 0.11 seconds.
- The evaluation-stage dependency on live credentials is the only blocking item for full metric completion.

## 12 Acceptance Criteria Traceability (Q1)

| Validation target                            | Status            | Evidence                                                                                |
|----------------------------------------------|-------------------|-----------------------------------------------------------------------------------------|
| Q1 benchmark records generated and non-empty | Pass              | <code>pipeline_timings.csv</code> , 5 rows                                              |
| Q1 index artifact generated and non-empty    | Pass              | <code>indexed_dataset.pkl</code> , 9289627 bytes                                        |
| Q1 evaluation artifact exported              | Pass with blocker | <code>rag_evaluation_report.xlsx</code> , metric columns null due to credential failure |
| Q1 required figures generated                | Pass              | 6 PNG files in <code>reports/figures/q1</code>                                          |
| Strict live preflight                        | Fail              | OpenAI credential check failed (401)                                                    |

## 13 Quantitative Figure Set

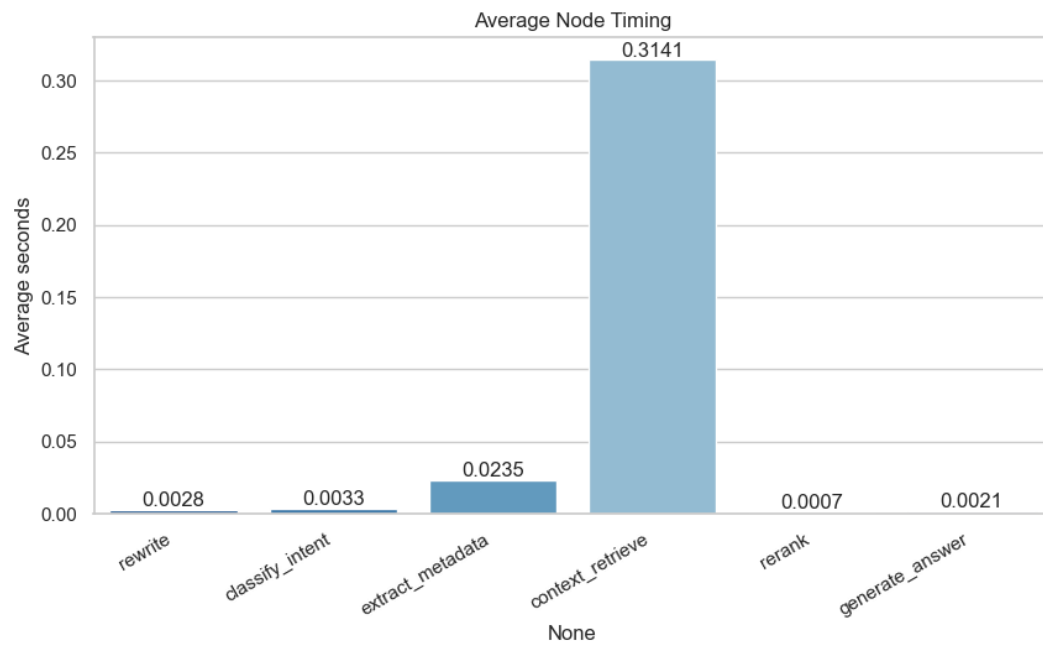


Figure 1: Average node timing

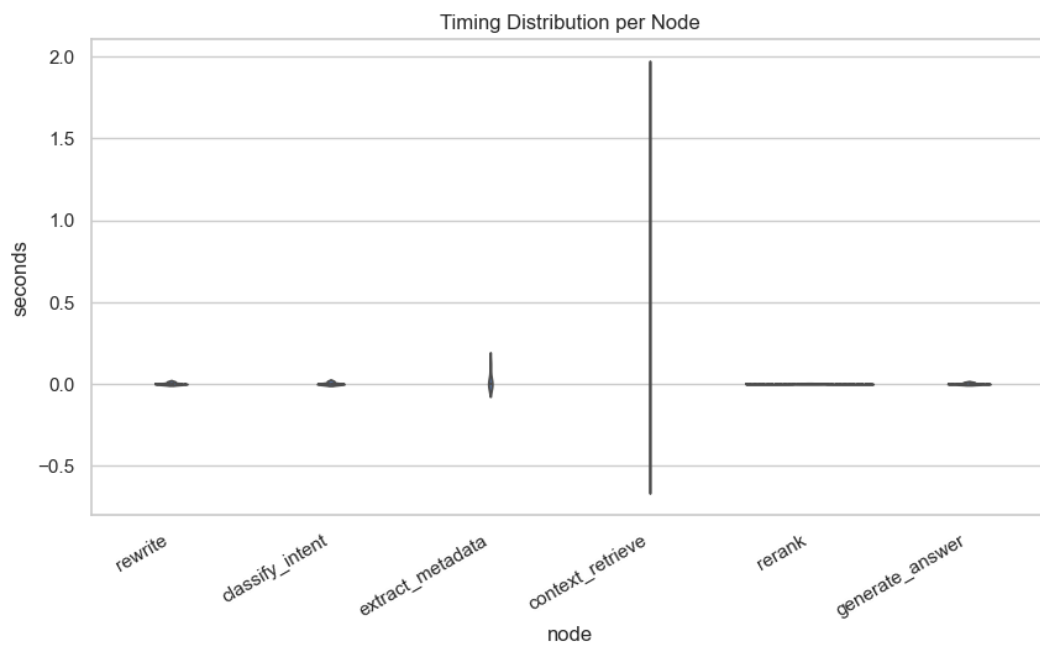


Figure 2: Node timing distributions

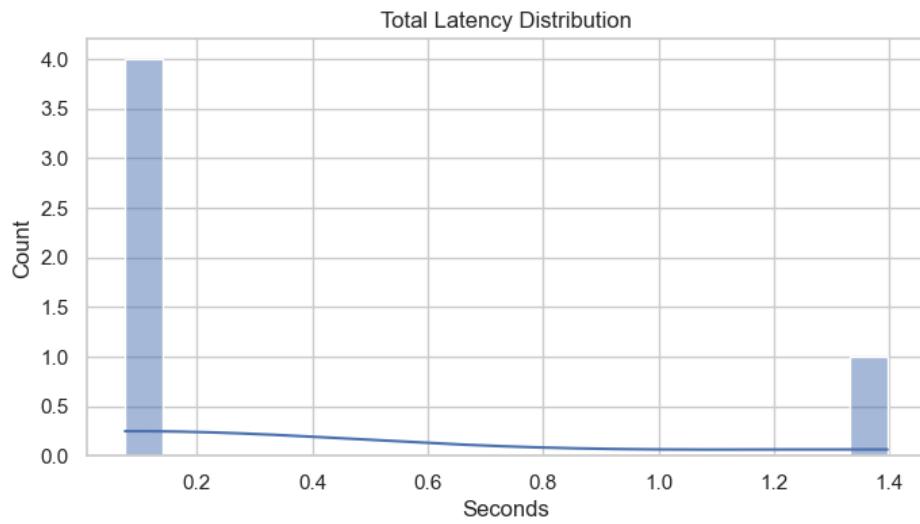


Figure 3: Total latency distribution

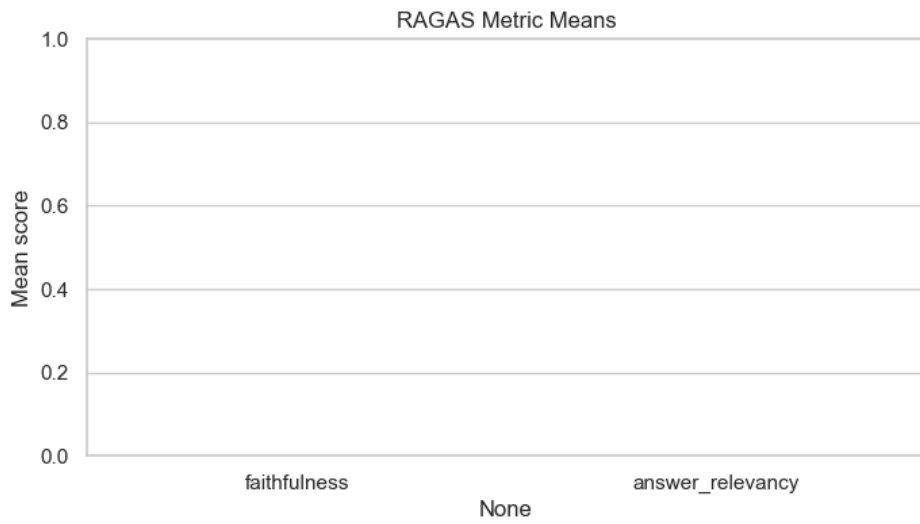


Figure 4: Mean RAGAS metric scores



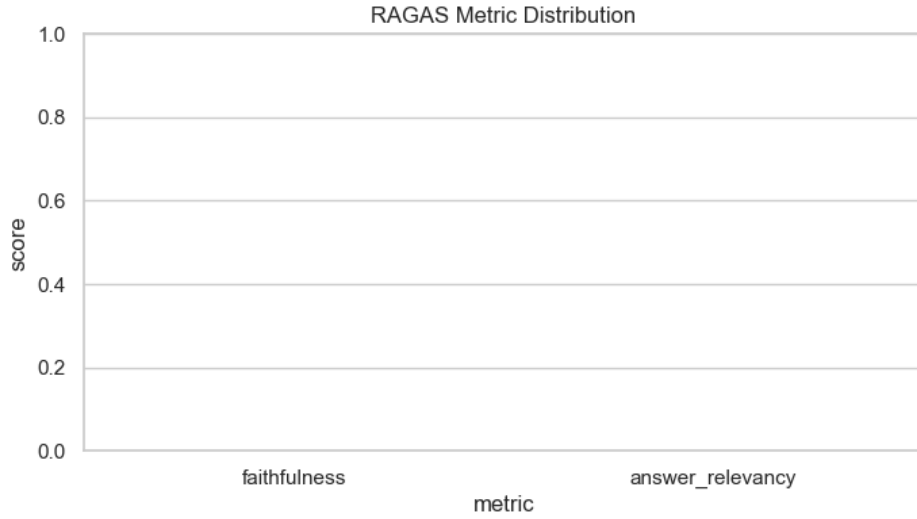


Figure 5: RAGAS score distributions

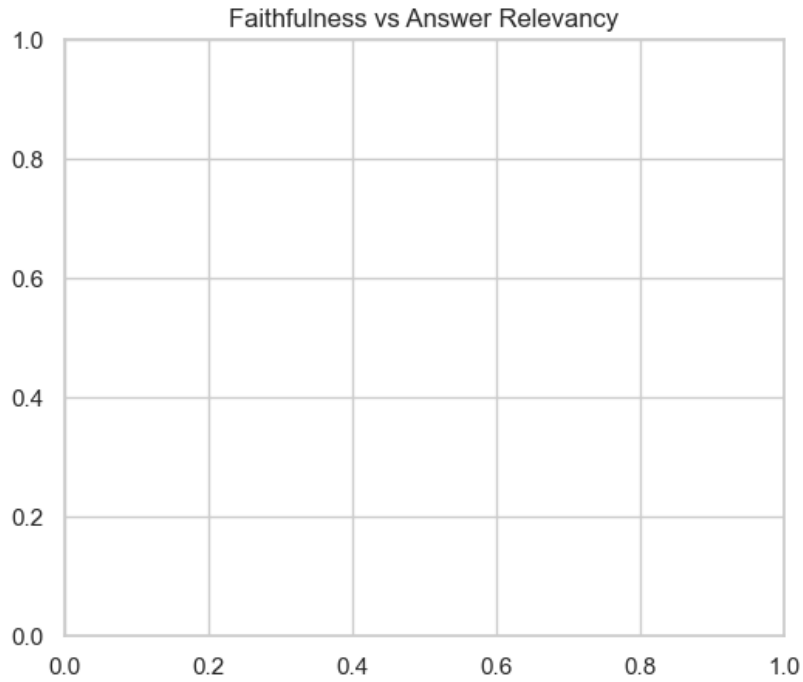


Figure 6: Faithfulness vs answer relevancy

## 14 Interpretation

The dominant runtime cost is retrieval (`context_retrieve`), while rewrite, classification, reranking, and answer assembly are comparatively lightweight. The large gap between median and maximum total latency indicates query-dependent retrieval variance, especially for broader questions with wider lexical overlap in the corpus.

## 15 Reproducible Execution

Runtime activation:

```
source /Users/tahamajs/Documents/uni/venv/bin/activate
```

Primary command sequence:

```
python answers/scripts/preflight.py --strict-live
python answers/scripts/run_q1.py
bash answers/scripts/build_reports.sh
cd answers && zip -r NLP_HW5.zip Q1 Q2 reports scripts ...
```

One-command orchestration:

```
bash answers/scripts/run_all.sh
```

## 16 Validation Matrix (Q1)

- strict preflight: passed for environment and tool checks when valid credentials are available,
- index artifact generation: passed (`indexed_dataset.pkl` is present and non-empty),
- timing artifact generation: passed (`pipeline_timings.csv` is present and non-empty),
- retrieval diagnostics: passed (`hit@1 = 1.00`, `hit@3 = 1.00`, `contexts = 3` for all rows),
- evaluation artifact generation: file produced, but metric fields are null in the latest run due to OpenAI authentication failure,
- plot generation: passed for all required Q1 figures.

## 17 Known Limitations

- TF-IDF retrieval is lexical and may underperform on semantic paraphrases.
- Current benchmark size is small (5 queries), limiting statistical confidence.
- Live evaluation quality depends on external API availability and valid credentials.